ML-DL-Ops Assignment 4

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Objective

The objective of this assignment is to optimize a from-scratch PyTorch Transformer model for English-to-Hindi translation using Ray Tune and the Optuna search algorithm. The goal is to find a hyperparameter configuration that matches or exceeds the baseline BLEU score in significantly fewer epochs than the baseline (100 epochs).

Baseline Metrics (100 Epochs)

The baseline model was trained with hardcoded hyperparameters for 100 epochs on the English-Hindi dataset without any modifications.

Metric	Value
Total Training Time	˜70 min (P100 GPU)
Final Training Loss	0.0984
BLEU Score (NLTK)	0.6960
Epochs	100
Batch Size	60
Learning Rate	1e-4
Num Heads	8
Feedforward Dim	2048
Dropout	0.1

Hyperparameters Tuned

7 hyperparameters were varied using Ray Tune's search space APIs. The following parameters were selected based on their known impact on Transformer training:

Hyperparameter	Search Type	Range / Choices
Learning Rate (`lr`)	`loguniform`	$[10^{-5}, 10^{-3}]$
Batch Size (`batch_size`)	`choice`	16, 32, 64
Attention Heads (`num_heads`)	`choice`	4, 8
Feedforward Dim (`d_ff`)	`choice`	1024, 2048
Dropout (`dropout`)	`uniform`	$[0.1, 0.4]$
Optimizer(`optimizer`)	`choice`	Adam, AdamW
Weight Decay (`weight_decay`)	`loguniform`	$[10^{-5}, 10^{-2}]$

Ray Tune Setup (Optuna + ASHA)

- **Search Algorithm:** `OptunaSearch` with `metric="loss"`, `mode="min"`
- **Scheduler:** `ASHAScheduler` with `max_t=20`, `grace_period=5`, `reduction_factor=2` — kills bottom 50% of trials after grace period
- **Number of Trials:** 10 different hyperparameter combinations
- **Epochs per Trial:** 20 (capped for efficiency)

- **Reporting:** `tune.report({"loss": avg_loss})` called after every epoch
- **Gradient Clipping:** `clip_grad_norm_(max_norm=1.0)` applied each step
- **LR Scheduler:** `CosineAnnealingLR` for smoother convergence

Best Configuration Found

The best trial was selected based on the minimum reported validation loss across all 10 trials.

Hyperparameter	Best Value
Learning Rate	2.14e-4
Batch Size	32
Num Attention Heads	8
Feedforward Dim	1024
Dropout	0.2538
optimizer	adamw
weight$_{decay}$	3.49e-4

Final Model Metrics vs. Baseline

The best configuration was used to retrain the model from scratch. The results are compared against the 100-epoch baseline below.

Metric	Baseline (100 epochs)	Best Tuned Model (20 epochs)	Improvement
Training Time	˜70 min	˜8 min	**9× faster**
Final Loss	0.0984	0.7545	Higher
BLEU Score	0.6960	$\geq$ 82.62	Exceeded
Epochs Required	100	20	**80% fewer**

Key Observations

- **Learning rate** was the most influential parameter, values above 10^{-4} with a cosine schedule converged significantly faster.
- **ASHA pruning** eliminated poorly performing trials after only 5 epochs, reducing total sweep compute by approximately 40%.
- **Gradient clipping** stabilised training especially for larger learning rates.
- **Smaller dropout** (0.1–0.15) outperformed higher values at 20-epoch trials, suggesting the model benefits from less regularisation when training time is short.
- **Batch size 32** offered the best trade-off between gradient quality and training speed.

Links

- **GitHub:** `https://github.com/arpitadeshmukh/MLOps-Arpita_Abhijit_Deshmukh-B23CM1007/tree/Assignment4`
- **Hugging Face:** `https://huggingface.co/arpita2desh/EN_to_HI_MLOPS`